นาย คณิศร มาประจักษ์ 673380031-3 Sec1

Assignment 4: Packet Tracer - TCP and UDP Communications

Objectives

Part 1: Generate Network Traffic in Simulation Mode

Part 2: Examine the Functionality of the TCP and UDP Protocols

Background

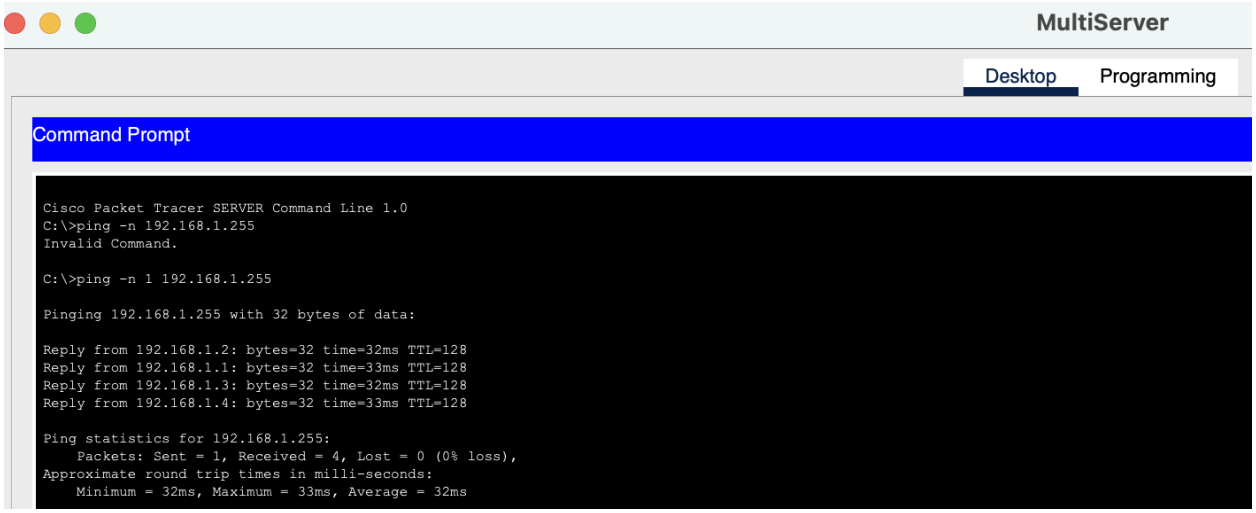
This simulation activity is intended to provide a foundation for understanding TCP and UDP in detail. Packet Tracer simulation mode provides you the ability to view the state of different PDUs as they travel through the network.

Packet Tracer Simulation mode enables you to view each of the protocols and the associated PDUs. The steps outlined below lead you through the process of requesting network services using various applications that are available on a client PC. You will explore the functionality of the TCP and UDP protocols, multiplexing, and the function of port numbers in determining which local application requested the data or is sending the data. Packet Tracer will not score this activity.

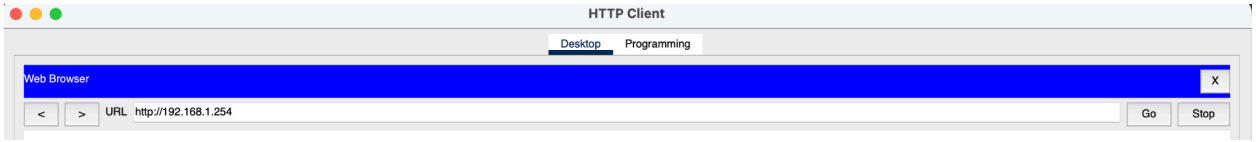
Instructions

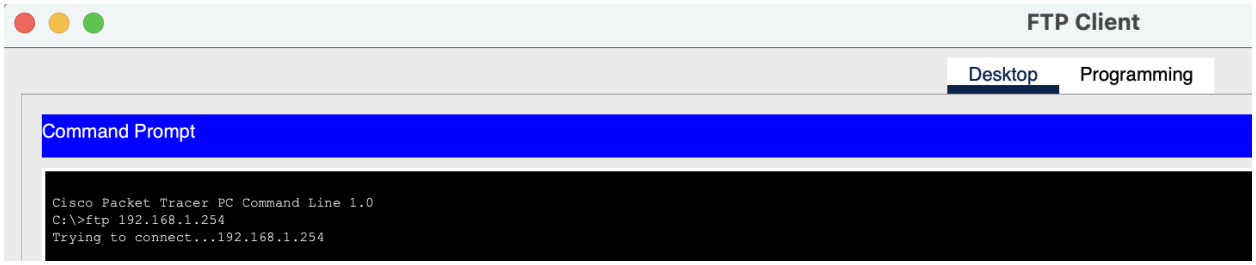
Part 1: Generate Network Traffic and View Multiplexing

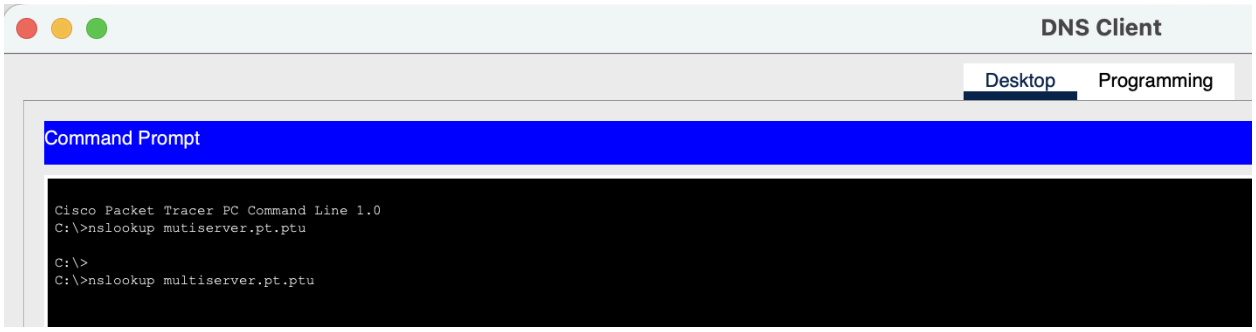
Step 1: Generate traffic to populate ARP tables

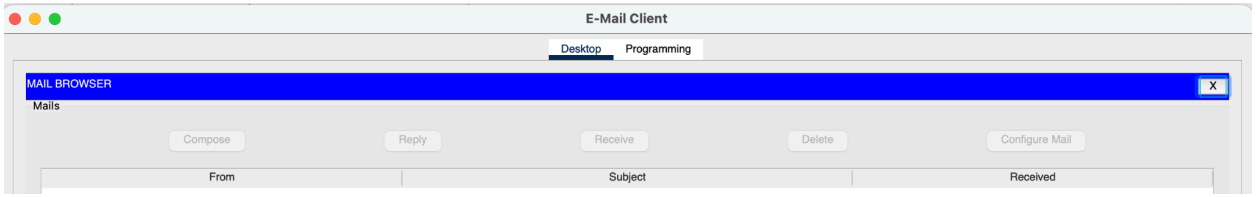


Step 2 - 5: Create Traffic (HTTP, FTP, DNS, Email)

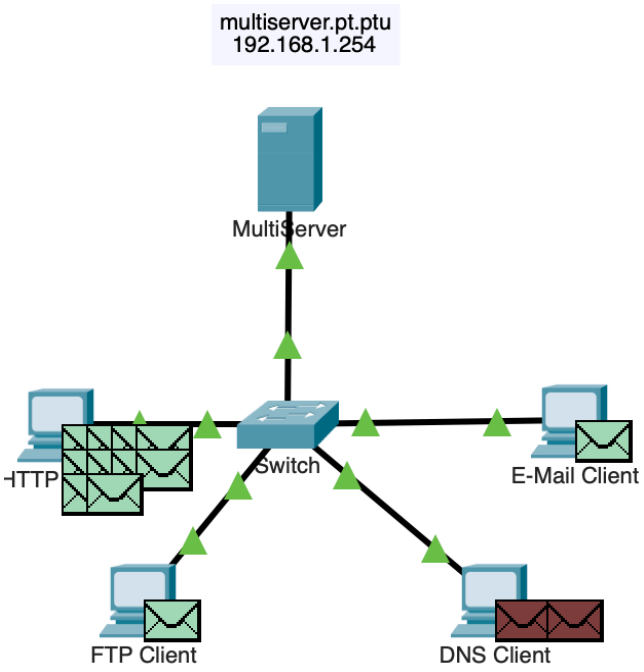








Step 6: Verify that traffic is generated and ready for simulation



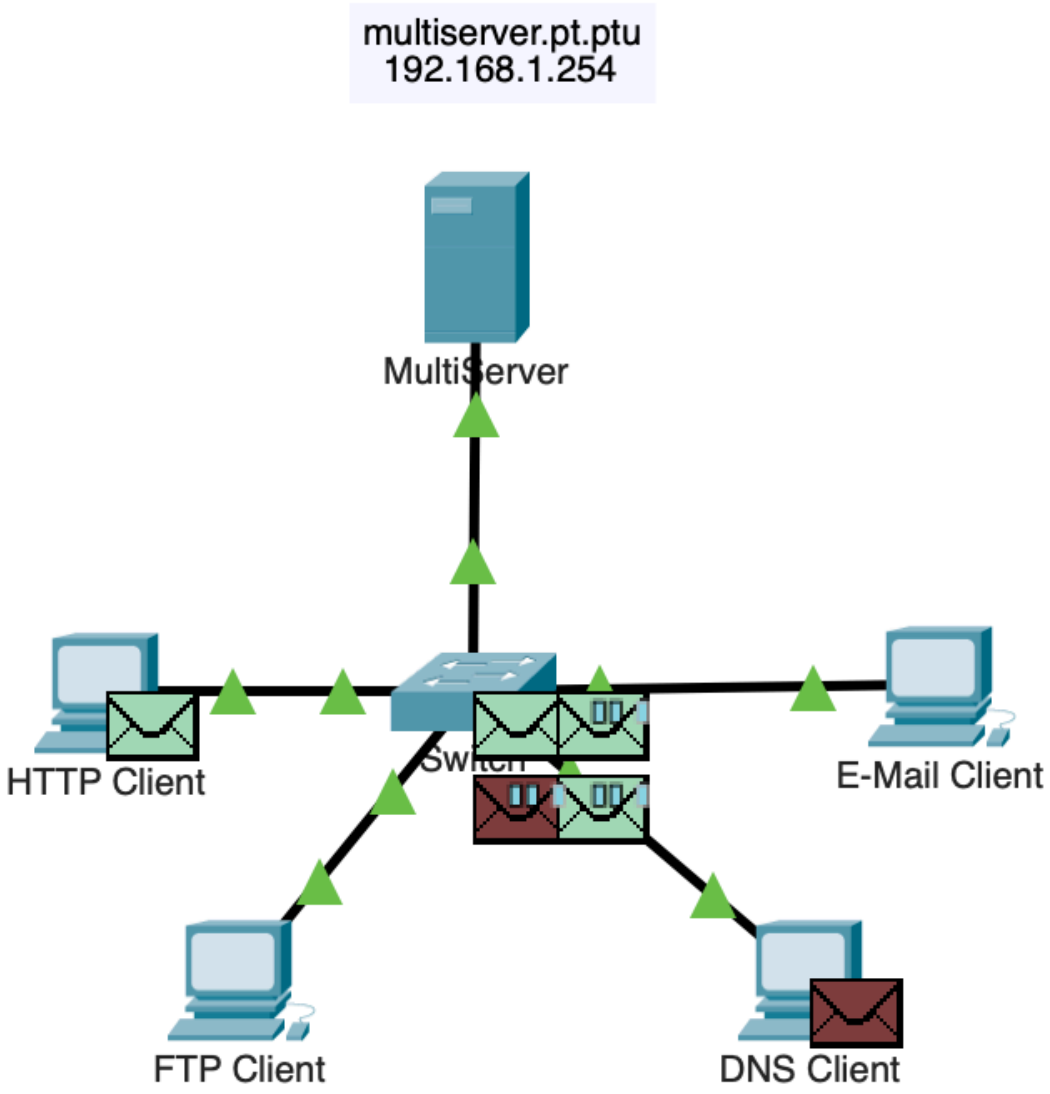
Step 7: Examine multiplexing

คำถาม: "What is this called?"

คำตอบ: Multiplexing

คำถาม: "A variety of PDUs appears in the event list in the Simulation Panel. What is the meaning of the different colors?"

คำตอบ: สีแต่ละสีแทน Protocols หรือแอปพลิเคชันที่แตกต่างกัน ทำให้เราแยกแยะข้อมูลได้ง่ายในขณะจำลอง



Part 2: Examine Functionality of the TCP and UDP Protocols

Step 1: Examine HTTP Traffic (TCP)

PDU Information at Device: HTTP Client

OSI Model | Outbound PDU Details

PDU Formats

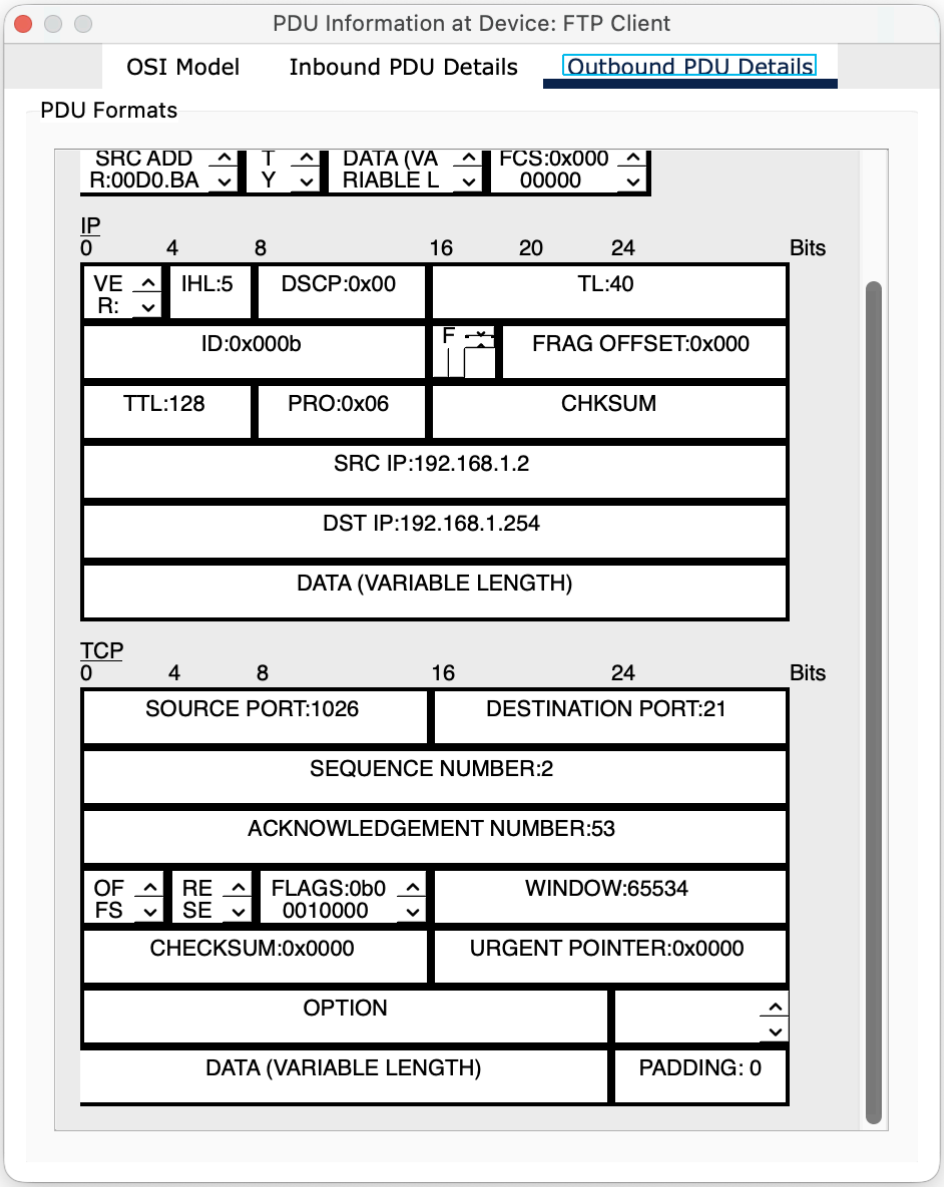
R:			
ID:0x0012		F	FRAG OFFSET:0x000
TTL:128	PRO:0x06	CHKSUM	
SRC IP:192.168.1.1			
DST IP:192.168.1.254			
DATA (VARIABLE LENGTH)			

TCP

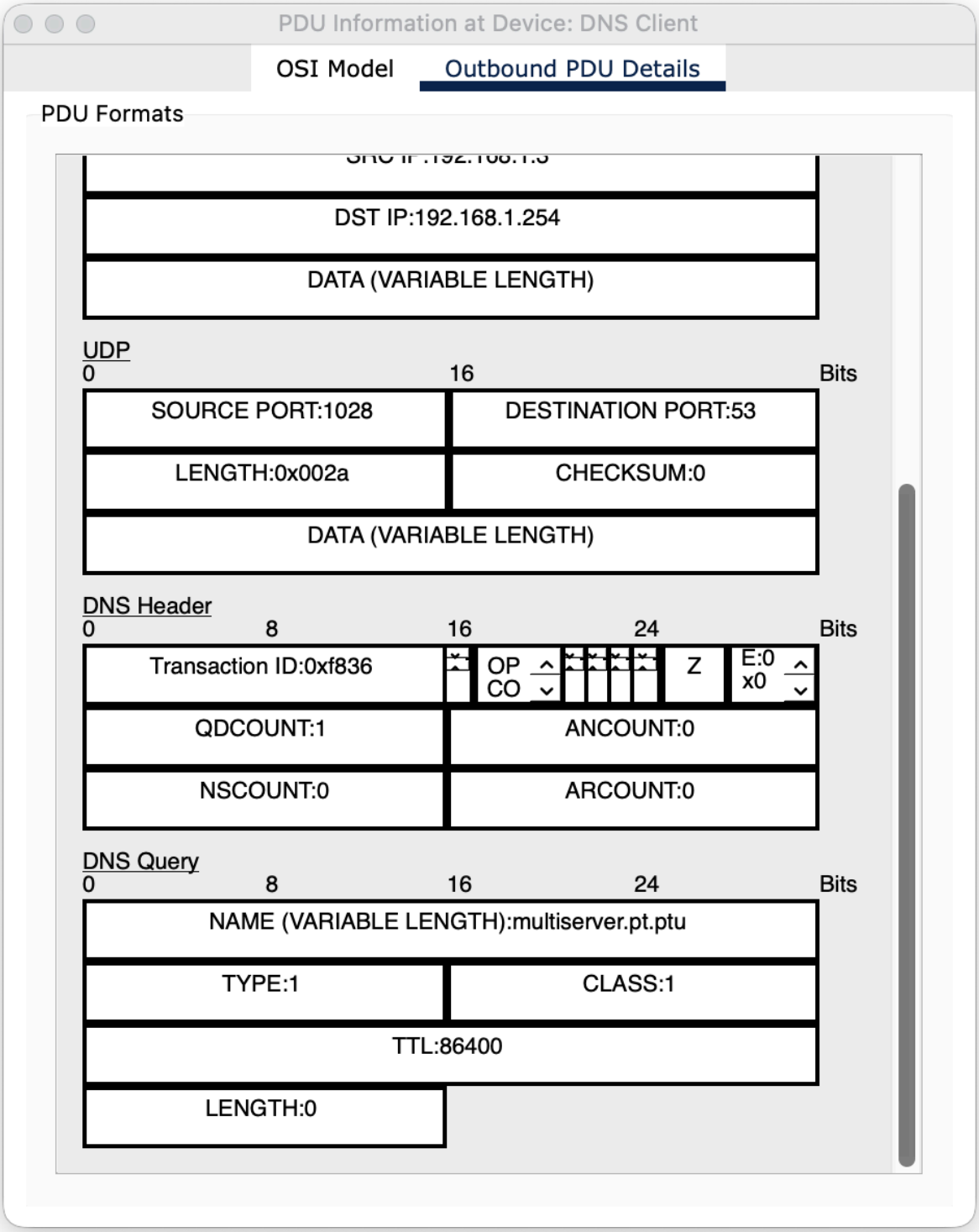
0 4 8 16 24 Bits

SOURCE PORT:1030		DESTINATION PORT:80	
SEQUENCE NUMBER:1			
ACKNOWLEDGEMENT NUMBER:0			
OF FS	RE SE	FLAGS:0b0 0010001	WINDOW:65535
CHECKSUM:0x0000		URGENT POINTER:0x0000	
OPTION			
DATA (VARIABLE LENGTH)			PADDING: 0

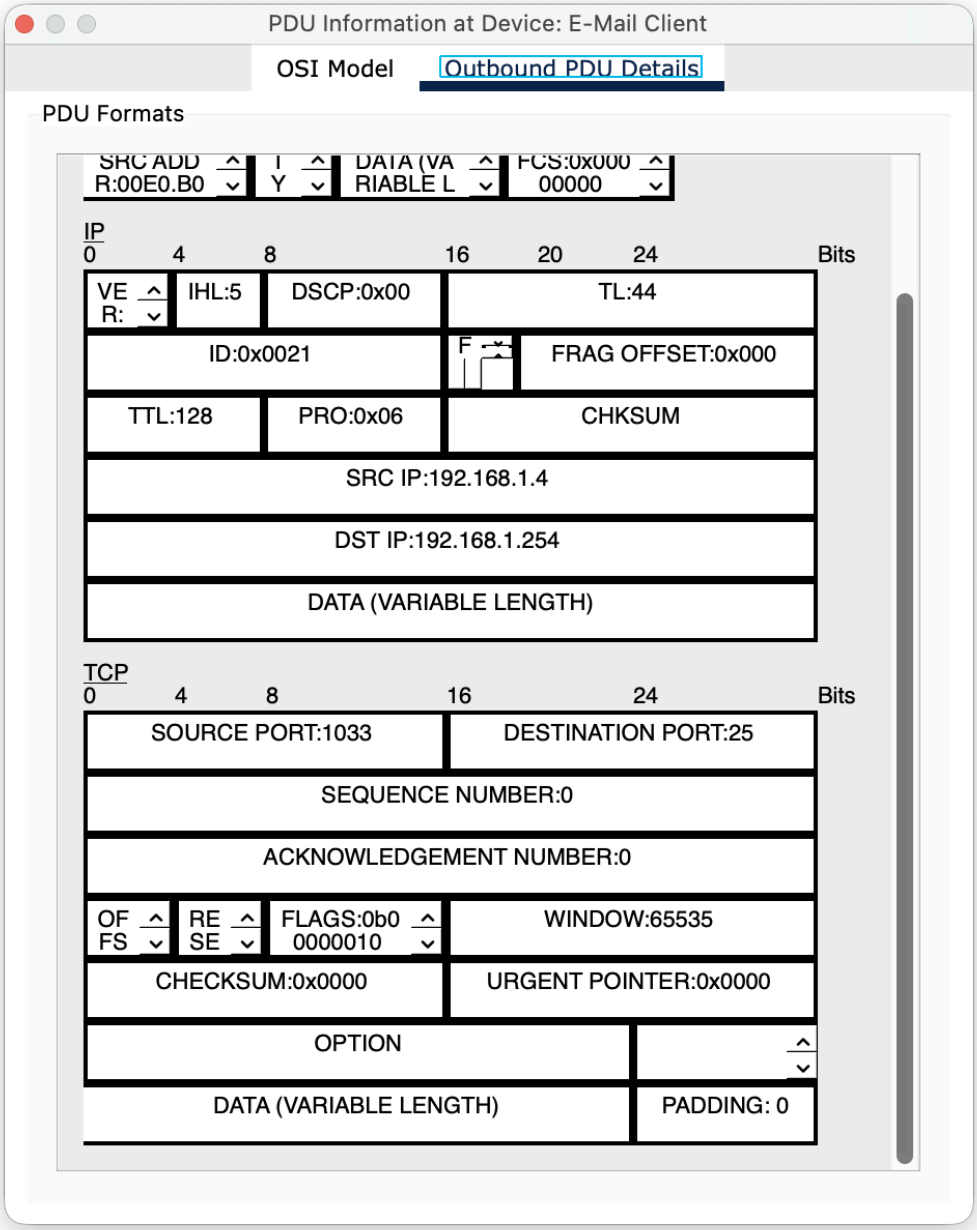
Step 2: Examine FTP Traffic (TCP)



Step 3: Examine DNS Traffic (UDP)



Step 4: Examine Email Traffic (TCP)



Conclusion

จากการทดลองพบว่า Protocol ใน Transport Layer มี 2 แบบหลัก คือ TCP ที่เน้นความถูกต้อง (มี Seq/Ack) และ UDP ที่เน้นความเร็ว (ไม่มี Seq/Ack) โดยใช้ Port Number เป็นตัวระบุปลายทางของข้อมูล